

Level 2.5 - Detect ARP poisoning (Static definition)

Attackers can use an ARP poisoning to impersonate another IP in the network.

You only need to protect against poisoning of the server's IP.

In this level we will detect malicious ARP responses by maintaining two static definitions in the code: the IP and MAC of the server.

Password

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Instructions

1. In your code, define both the server's IP and MAC
2. Simple - alert on any ARP response that claims to know the server's IP, but the MAC isn't what's expected
3. To trigger the attack
 - Run the attacker's malicious trigger for level 2.5 - this will start the ARP poisoning (You may stop it at any point with the "Stop" button)
 - Only then, run the client's non-malicious trigger for level 2.5 - it should be poisoned by the attacker.
4. In your alert, include the victim's IP, the poisoned IP (the server's IP), and the attacker's MAC

Notes

Use the guiding questions below to identify the malicious trigger and structure your code's detection:

- Which packets do we need to handle?
 - Protocol?
 - Request or Response?
- How do we compare the static definitions?

To submit

Submit a ZIP `nids.zip` containing all `.py` files.

